Post Graduate Program in Data Science with Generative AI

Decision Making with Business Insights
Business Report

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Problem 1:

A physiotherapist with a male football team is interested in studying the relationship between foot injuries and the positions at which the players play from the data collected.

	Striker	**Forward**	**Attacking Midfielder**	**Winger**	**Total**
Players Injured	45	56	24	20	145
Players not Injured	32	38	11	9	90
Total	77	94	35	29	235

1.1 What is the probability that a randomly chosen player would suffer an injury?

Solution:

Total number of players = 235

Number of players injured = 145

Probability of Injury = Numbers of players injured / Total number of players

Probability of Injury = 145 / 235 = 0.617

Probability that a randomly chosen player would suffer an injury = 61.7 %

1.2 What is the probability that a player is a forward or a winger?

Solution: Number of forwards = 94

Number of wingers = 29

Probability of being a forward or a winger = (No of forwards + No of wingers) / Total number of players

Probability of being a forward or a winger = (94 + 29) / 235 = (123 / 235) = 0.5234

The probability that a player is a forward or a winger is 52.34 %.

1.3 What is the probability that a randomly chosen player plays in a striker position and has foot injury?

Solution:

Number of strikers with foot injury = 45

Probability of being a striker with foot injury = Number of strikers with foot injury / Total number of players.

Probability of being a striker with foot injury = 45 / 235 = 0.191

The probability that a randomly chosen player plays in a striker position and has a foot injury is 19.1%.

1.4 What is the probability that a randomly chosen injured player is either a forward or an attacking midfielder?

Solution:

Number of forwards injured = 56

Number of attacking midfielders injured = 24

Probability of being a forward or an attacking midfielder given injury = (Number of forwards injured + Number of attacking midfielders injured) / Number of injured players

Probability of being a forward or an attacking midfielder given injury = (56 + 24) / 145 = 0.5517

The probability that a randomly chosen injured player is either a forward or an attacking midfielder is 55.17 %.

Problem 2:

The breaking strength of gunny bags used for packaging cement is normally distributed with a mean of 5kg per sq. centimetre and a standard deviation of 1.5 kg per sq. centimetre. The quality team of the cement company wants to know the following about the packaging material to better understand wastage or pilferage within the supply chain; Answer the questions below based on the given information; (Provide an appropriate visual representation of your answers, without which marks will be deducted).

2.1 What proportion of the gunny bags have a breaking strength less than 3.17 kg per sq.cm?

Solution:

The cumulative distribution function indicates that 11.12% of gunny bags have a breaking strength below 3.17 kg per square centimeter.

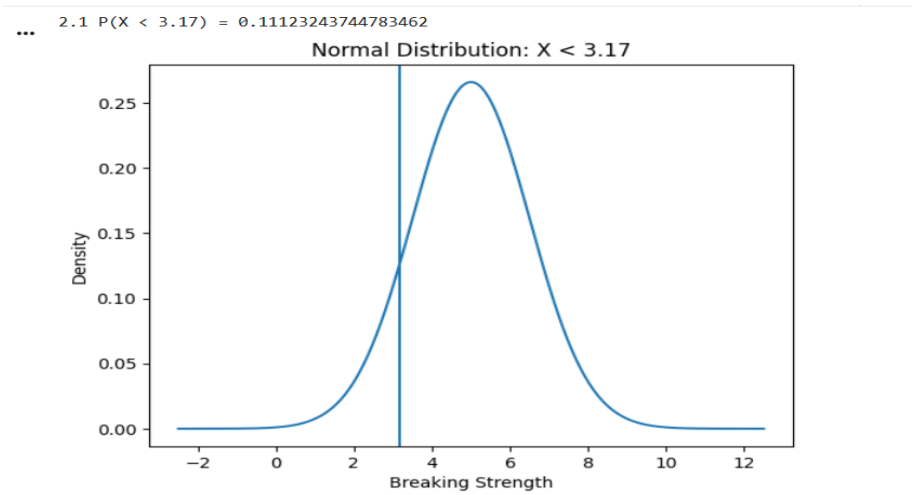


Fig. 2.1 breaking strength less than 3.17 kg per sq.cm

2.2 What proportion of the gunny bags have a breaking strength at least 3.6 kg per sq.cm?

Solution:

An assessment using the cumulative distribution function indicates that 82.46% of the gunny bags exhibit a breaking strength of no less than 3.6 kg per square centimeter.

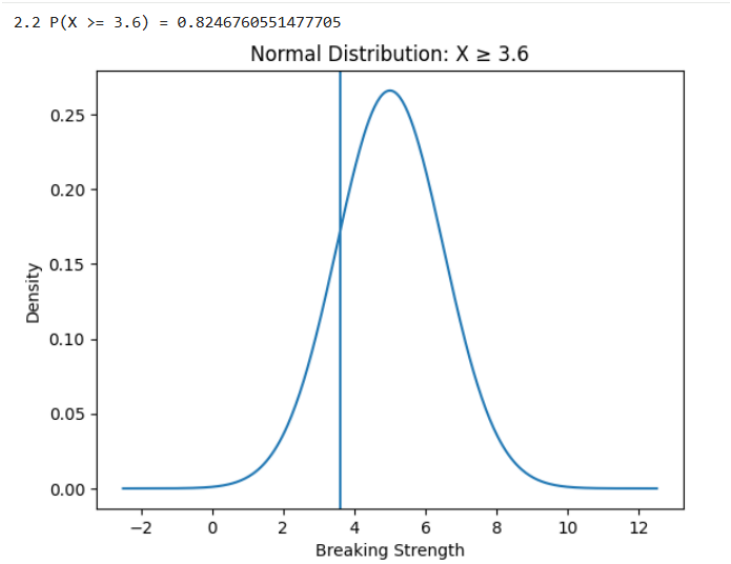


Fig: 2.2 breaking strength at least 3.6 kg per sq.cm

2.3 What proportion of the gunny bags have a breaking strength between 5 and 5.5 kg per sq.cm.?

Solution:

Based on the cumulative distribution function, 13.06% of the gunny bags have a breaking strength in the range of 5 to 5.5 kg per square centimeter.

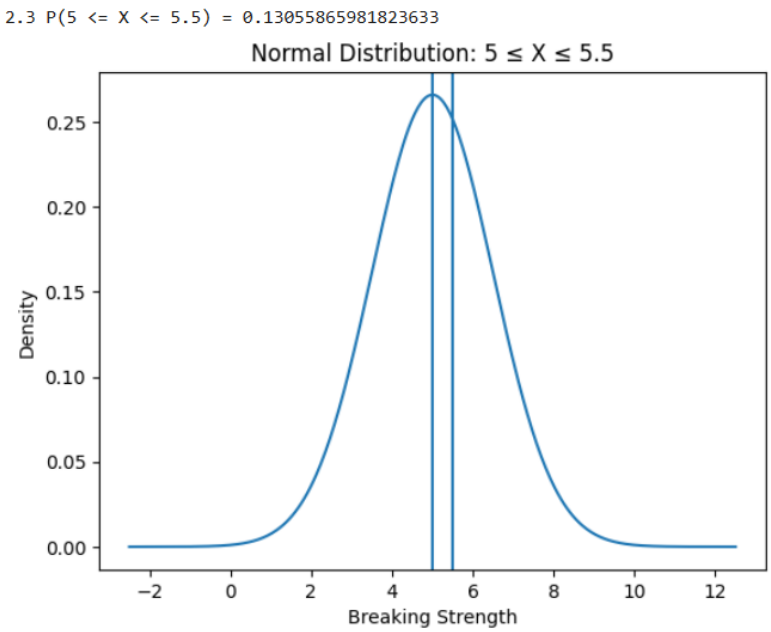


Fig 2.3 breaking strength between 5 and 5.5 kg per sq.cm.

2.4 What proportion of the gunny bags have a breaking strength NOT between 3 and 7.5 kg per sq.cm.?

Using the cumulative distribution function, it is determined that 14% of the gunny bags have a breaking strength outside the range of 3 to 7.5 kg per square centimetre.

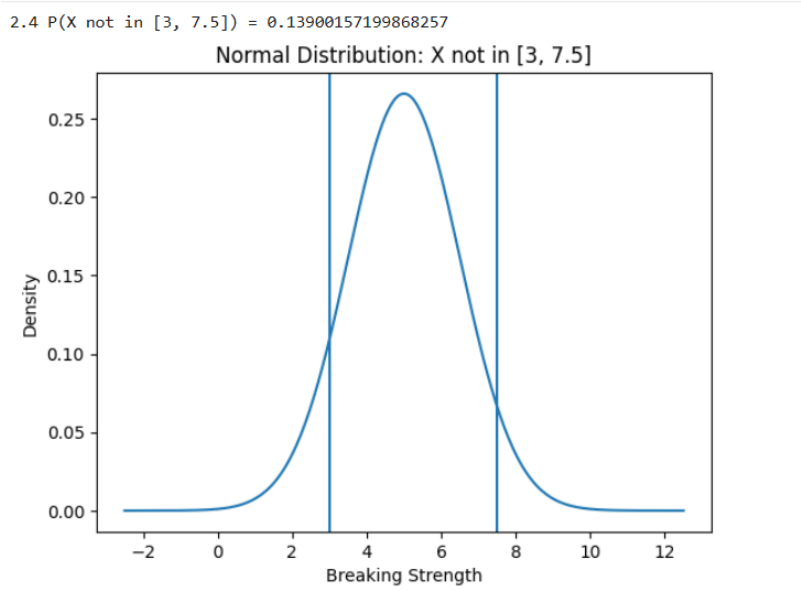


Fig 2.4 breaking strength NOT between 3 and 7.5 kg per sq.cm

<u>Problem 3:</u>

Zingaro stone printing is a company that specializes in printing images or patterns on polished or unpolished stones. However, for the optimum level of printing of the image the stone surface has to have a Brinell's hardness index of at least 150. Recently, Zingaro has received a batch of polished and unpolished stones from its clients. Use the data provided to answer the following (assuming a 5% significance level);

3.1 Zingaro has reason to believe that the unpolished stones may not be suitable for printing. Do you think Zingaro is justified in thinking so?

<u>Solution:</u>

- **Null Hypothesis (H_0):** The mean hardness of unpolished stones is 150 ($\mu = 150$).
- **Alternative Hypothesis (H_1):** The mean hardness of unpolished stones is less than 150 ($\mu < 150$).
- **Hypothesis Test:** A one-tailed t-test was conducted at a 5% significance level.

  Test statistic: $t = -4.16$

  p-value = 4.17×10^{-5}
- **Conclusion:** Since the p-value is less than 0.05, the null hypothesis is rejected. There is sufficient evidence to conclude that the mean hardness of the unpolished stones is less than 150. Therefore, Zingaro is justified in believing that the stones may not be suitable for printing.

```
n = 75
unpolished_mean = 134.11052653373332
std_unpolished = 33.0418044136061

Test for unpolished < 150
t-statistic = -4.164629601426757
one-sided p-value = 4.171286997419652e-05
```

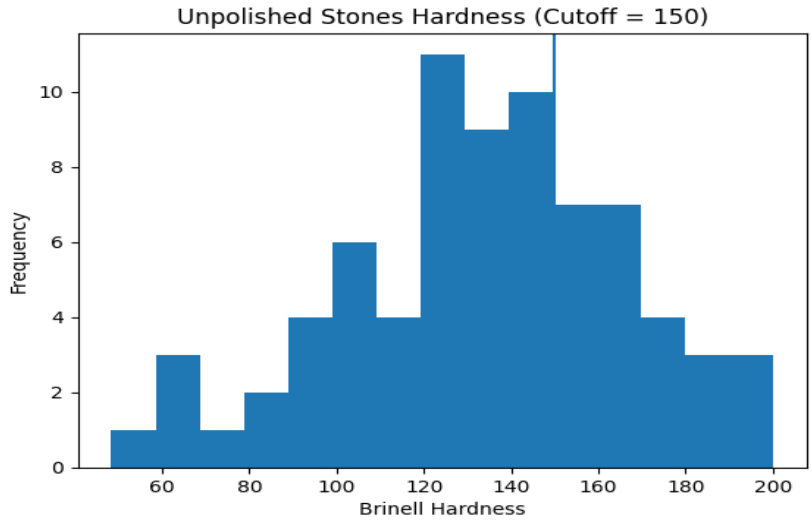


Fig 3.1 Unpolished stones hardness

3.2 Is the mean hardness of the polished and unpolished stones the same?

<u>Solution:</u>

- **Null Hypothesis (H_0):** The mean hardness of polished and unpolished stones is equal ($\mu_1 = \mu_2$).
- **Alternative Hypothesis (H_1):** The mean hardness of polished and unpolished stones is not equal ($\mu_1 \neq \mu_2$).
- **Hypothesis Test:** A two-tailed t-test was conducted to compare the means. The test statistic was 3.55 and the p-value was 0.00065.
- **Conclusion:** Since the p-value (0.00065) is less than the significance level of 0.05, we reject the null hypothesis. This indicates a significant difference in mean hardness between polished and unpolished stones.

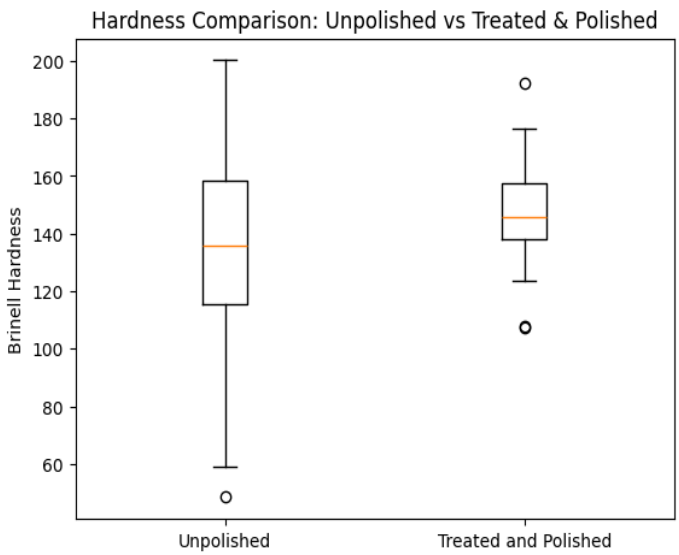


Fig 3.2 Hardness comparison

Problem 4:

Dental implant data: The hardness of metal implants in dental cavities depends on multiple factors, such as the method of implant, the temperature at which the metal is treated, the alloy used as well as the dentists who may favour one method above another and may work better in his/her favourite method. The response is the variable of interest.

4.1 How does the hardness of implants vary depending on dentists?

Solution:

For Alloy 1

Hypotheses:

- H_{01}: Mean hardness is the same for all dentists (Alloy 1).
- H_{11}: At least one dentist has a different mean hardness (Alloy 1).

Test: One-Way ANOVA

Result: p-value = 0.1166

Decision: p-value > 0.05 → Fail to reject H_{01}

Conclusion: There is no significant difference in implant hardness among dentists for Alloy 1.

For Alloy 2

Hypotheses:

- H_{02}: Mean hardness is the same for all dentists (Alloy 2).
- H_{12}: At least one dentist has a different mean hardness (Alloy 2).

Test: One-Way ANOVA

Result: p-value = 0.7180

Decision: p-value > 0.05 → Fail to reject H_{02}

Conclusion: There is no significant difference in implant hardness among dentists for Alloy 2

Post-hoc Test

Since ANOVA is **not significant** for both alloys, **no pairwise (post-hoc) test is required**.

Conclusion: At the 5% significance level, implant hardness does not differ significantly among dentists for either Alloy 1 or Alloy 2.

4.2 How does the hardness of implants vary depending on methods?

Solution:

Hypotheses:

- H_0: Mean implant hardness is the same for all methods.
- H_1: At least one method has a different mean hardness.

Assumptions:

- Normality checked using Shapiro–Wilk test.
- Homogeneity of variances checked using Levene's test.
- As instructed, ANOVA is performed even if assumptions are violated.

Test Used: One-way ANOVA (Factor: Method), separately for each alloy.

Results:

- Alloy 1: p-value < 0.05 → Reject H_0.
- Alloy 2: p-value < 0.05 → Reject H_0.

Conclusion: Implant hardness differs significantly among methods for both alloys.

Post-hoc Analysis (Tukey HSD):

- **Alloy 1:**
  - Method 1 vs Method 2: Not significant
  - Method 1 vs Method 3: Significant difference
  - Method 2 vs Method 3: Significant difference
- **Alloy 2:**
  - Method 1 vs Method 2: Not significant
  - Method 1 vs Method 3: Significant difference
  - Method 2 vs Method 3: Significant difference

Final Interpretation:

Method 3 differs significantly from Methods 1 and 2 for both alloys, while Methods 1 and 2 do not differ significantly from each other.

4.3 What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?

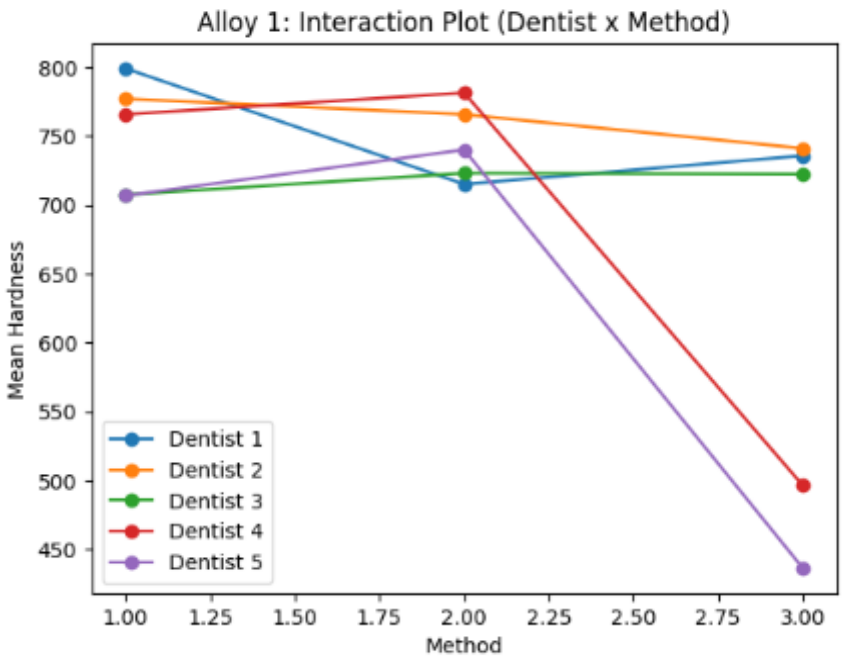
Solution:

Method: Interaction plots of mean hardness vs method for each dentist, plotted separately for Alloy 1 and Alloy 2.

Inference from Plots: The lines are not parallel, indicating interaction between Dentist and Method.

Conclusion: The effect of method on hardness depends on the dentist.

Final Statement: There is an interaction effect between Dentist and Method on implant hardness for both alloys.



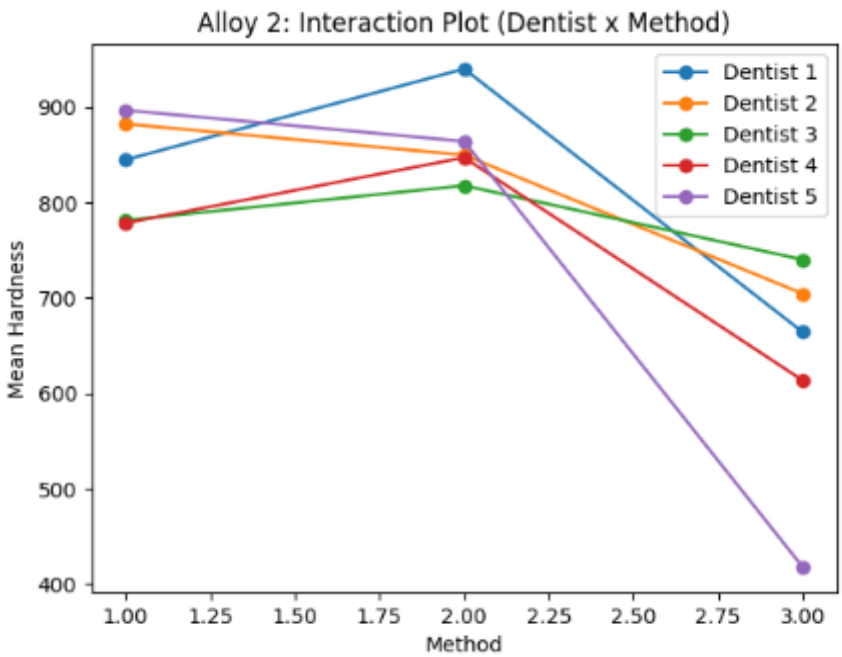


Fig 4.3 Interaction plot

4.4 How does the hardness of implants vary depending on dentists and methods together?

<u>Solution:</u>

Hypotheses:

- H_0(Dentist): Mean hardness is the same for all dentists.
- H_0(Method): Mean hardness is the same for all methods.
- H_0(Interaction): No interaction between dentist and method.
- H_1: At least one of these effects is significant.

Assumptions:

- Normality and equal variance assumptions were checked.
- As instructed, the test is conducted even if assumptions are violated.

Test Used: Two-way ANOVA with interaction, separately for each alloy.

Decision Rule ($\alpha = 0.05$):

If p-value < 0.05 → Reject corresponding H_0.

Results (from ANOVA table):

- Method: Significant.
- Dentist: Not significant.
- Interaction (Dentist × Method): Significant.

Conclusion:

- Implant hardness depends significantly on the method and on the interaction between dentist and method.
- The effect of method changes with dentist.

```
--- Alloy 1 ---
                          sum_sq    df          F    PR(>F)
C(Dentist)            106683.688889   4.0   3.899638  0.011484
C(Method)             148472.177778   2.0  10.854287  0.000284
C(Dentist):C(Method)  185941.377778   8.0   3.398383  0.006793
Residual              205180.000000  30.0        NaN       NaN

--- Alloy 2 ---
                          sum_sq    df          F    PR(>F)
C(Dentist)             56797.911111   4.0   1.106152  0.371833
C(Method)             499640.400000   2.0  19.461218  0.000004
C(Dentist):C(Method)  197459.822222   8.0   1.922787  0.093234
Residual              385104.666667  30.0        NaN       NaN
```

Fig 4.4 hardness of implants vary depending on dentists and methods